

RD900 locus structure project description

Background

We have previously published a paper detailing variation across the *Mycobacterium tuberculosis* Complex (MTBC) in the RD900 region. This region was first described as a lineage specific locus in *M. africanum* GM041182 strain (Bentley et al. 2012) and was thought to be deleted in *M. bovis* and “modern” *M. tuberculosis* lineages. RD900 region was not present in the original *M. bovis* AF2122/97 reference genome annotation, but was found to be actually present upon re-sequencing. These species show more than 99% genetic identity but exhibit distinct host preference and virulence. Differences in key loci are likely to be important in determining this.

The RD900 region comprises two homologous genes, *pknH1* and *pknH2*, encoding a serine/threonine protein kinase PknH flanking the *tbd2* gene. Our analysis revealed that RD900 has been independently lost in different MTBC lineages and different strains, resulting in the generation of a single *pknH* gene. Importantly, all the analysed *M. bovis* and *M. caprae* strains carry a conserved deletion within a proline rich-region of *pknH*, independent of the presence or absence of RD900. We hypothesized that deletion of *pknH* proline rich-region in *M. bovis* may affect PknH function, having a potential role in its virulence and evolutionary adaptation.

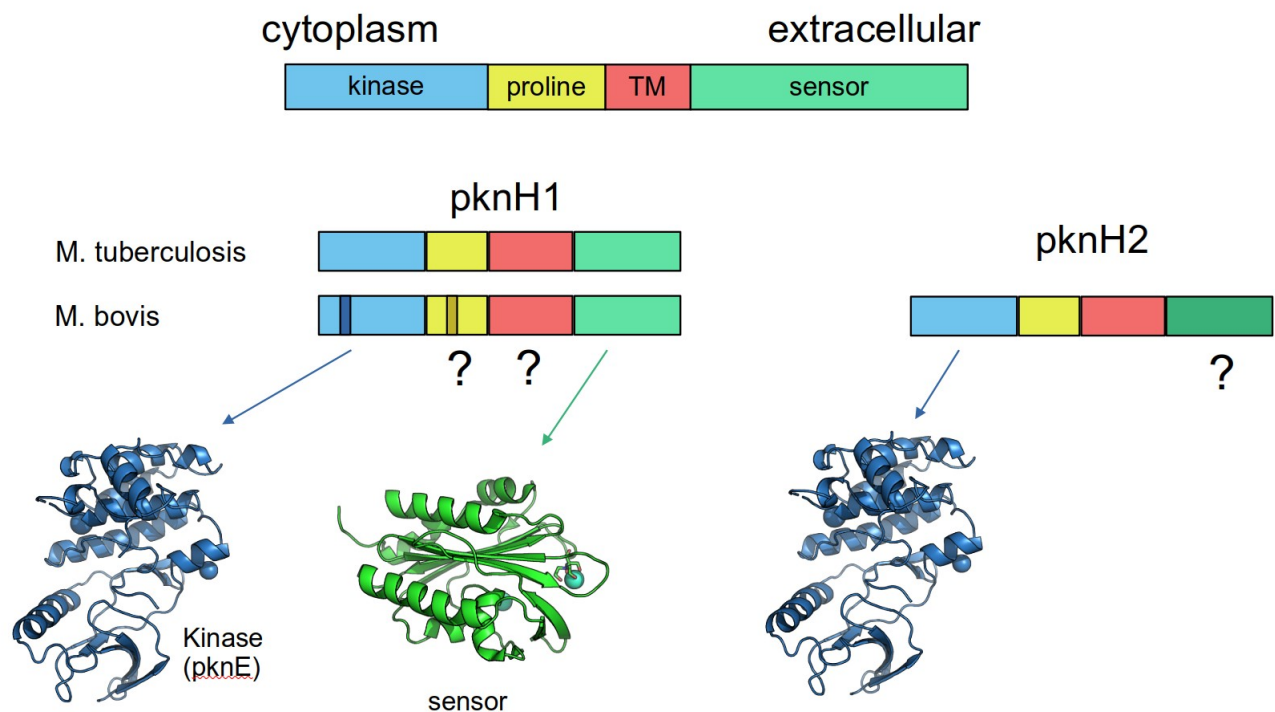


Figure 1: Structure of the *pknH1* and *pknH2* genes.

In this study we examined over 60 different MTBC strains representing lineages across the MTBC to see variations in the region. To accurately detect locus structure we used the raw read data instead of relying on assembled genomes which may have errors due to homology in the kinase domains of the *pknH* proteins, if both are present. Raw fastq files for all strains were downloaded and aligned to the

most complete version of the RD900 locus encoded in the *Mycobacterium africanum* (MAF) genome. This allowed missing features to be seen as gaps, as shown below.

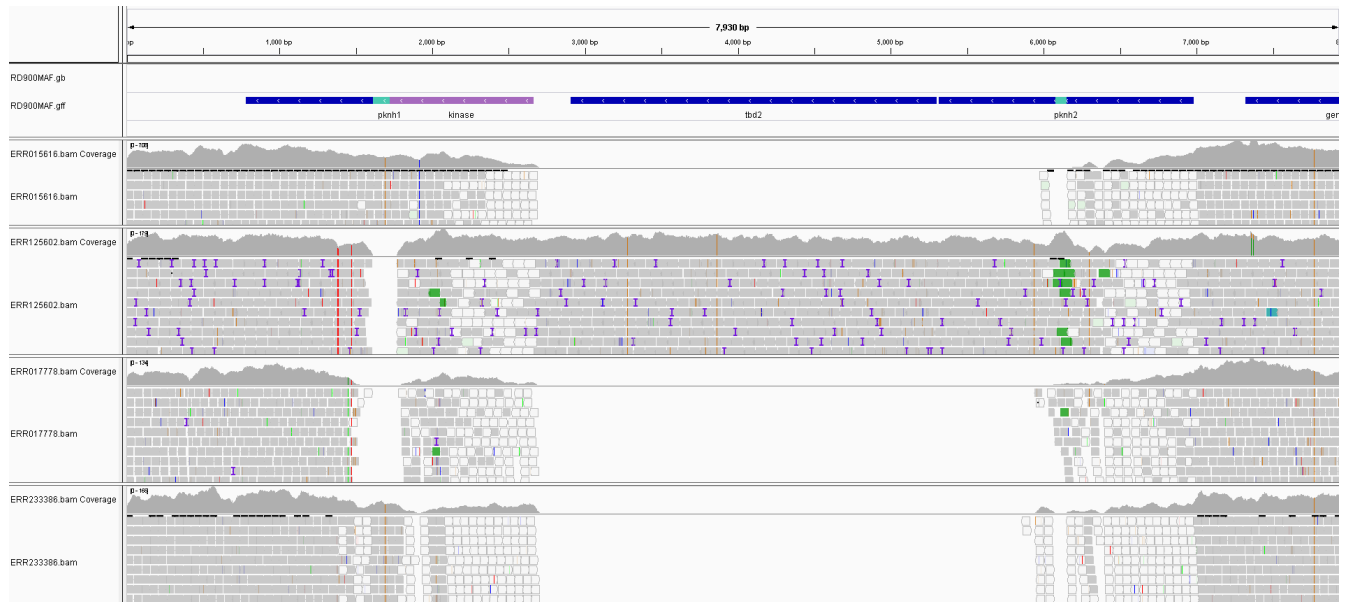


Figure 2: Alignments of some samples to the RD900 MAF region show the gaps where features are missing. The white reads are those multi-mapping to the homologous kinase domains of pknH1. This is what causes issues in assembly at this locus.

Objectives

Since this project was done in 2019, new data has been produced for all of these mycobacteria genomes including the animal adapted species we are most interested in. We will use this new data to provide a more complete analysis. As well as using raw read data as before we will use complete assemblies as a complementary approach. Complete assemblies should cover the whole locus and give a reliable picture of its structure.

We will:

- Retrieve new raw read post 2019 data to apply the same approach as before
- As a complementary approach, we will find all complete assemblies for the same set of species. In particular this may help find a more complete locus for *Africanum*, which we can align to.
- In both cases we need to reduce redundancy drastically to get a manageable dataset. I will handle this part.
- It might be that we can perform both methods in parallel and then compare results but one will feed into the other.

Methods

1. The assembly approach

There are >600 complete MTBC assemblies available. These have already been downloaded. What is needed prior to comparisons is to reduce the redundancy of these genomes. That is, to remove those that are too similar to each other.

Mash method for collapsing samples

Redundancy among genome assemblies can be reduced using Mash, a MinHash-based tool for estimating pairwise genomic distance without full alignment. For each assembly, a MinHash sketch is generated using Mash's default parameters (k-mer size $k = 21$, sketch size $s = 1,000$), and pairwise Mash distances – average nucleotide identity (ANI) - are computed for all assembly pairs (mash sketch followed by an all-vs-all mash dist).

Near identical genomes are clustered by single-linkage below a fixed threshold (0.001, corresponding to approximately >99.99% estimated ANI). For each cluster, a single representative assembly is retained – e.g. the member with the greatest total assembly length - and all other cluster members can be excluded from downstream analysis.

On threshold: 0.001 is a good starting point for collapsing near-duplicate/outbreak-level redundancy (>99.9% similarity). But use different thresholds to see the results.

Find lineages

This will be done with a tool called TB-profiler. I will do this part and add it to the table.

Extracting the RD900 locus

We chose to adopt an anchor based extraction + alignment method. We find the conserved anchor areas around the locus first, then **BLAST the two flanking anchor genes separately** (not the internal content), then take whatever lies between their hit coordinates in each assembly as "the locus," regardless of what's in there. This sidesteps the paralog-confusion problem and should correctly extract the interval *whatever its content*: intact, deleted, partial, or bearing a large insertion, with no assumption about what should be inside.

Comments/caveats on this approach in general:

- Anchor choice matters more than anything else. They need to be genuinely unique and conserved across *all* species in - so it's worth explicitly checking copy number of your chosen anchor genes across the assemblies.
- Assembly fragmentation breaks it silently. But we are using complete genomes so that should be ok, but do not assume 100% extraction across all assemblies.

Compare the RD900 sequences

We can do a multiple sequence alignment of the locus for all samples. Using mafft is fastest.

We can also build a phylogeny with *fasttree* to get an idea of how the locus sequences line up as compared to their species relationship.

The most useful method for comparison may be to annotate the sequences (e.g. with *prokka*) and compare them with clinker.

Steps in detail

Here are some of the commands for the assembly analysis. Your filenames may differ in some cases.

Extract the up and downstream anchors:

Here is the link to the genome viewer for the reference Africanum genome where you can extract the flanking sequences from the region.

<https://www.ncbi.nlm.nih.gov/nuccore/FR878060.1?report=graph&from=1411192&to=1416081>

Use mash to get all to all similarity and cluster/reduce genomes. This produces a new folder with the reduced genomes. This is all done in the script below. You can change the threshold if needed.

```
python mash_reduce.py --assemblies mtb_genomes --threshold 0.0005
```

The script locus_extraction.py will then extract the locus sequences:

```
python locus_extraction.py --up upstream_anchor.fa --down downstream_anchor.fa --assemblies mtb_reduced/
```

Align the sequences:

```
mafft --auto --thread 4 RD900_extracted.fa > RD900_extracted_aligned.fa
```

Make a phylogeny from the alignment:

```
fasttree -nt -gtr RD900_extracted_aligned.fa > RD900_locus_tree.nwk
```

We can also try to look at the locus structures and compare them by annotating each locus and then viewing with *clinker*. The script annotate.py will identify all the proteins in each locus and create a gbk file.

```
python annotate.py -p RD900MAF.faa -o annot RD900_extracted.fa
```

```
clinker annot/*.gbk -o clinker_alignment.html -p clinker_plot.html -i 0.75
```

Then view the result in clinker. We will use this to distinguish locus structures and correlate this with lineage.

2. Raw reads alignment approach

This is the method we used before. It avoids the issues of erroneous assembly which may exist even in the complete ones. Especially in regions like RD900. Also there is a larger raw read dataset (as not all raw WGS data is assembled and uploaded and most are never made into complete genomes).

We do not have the files yet so concentrate on the assemblies.

Datasets

On the shared drive there are now several sets of data:

mtbc_assemblies.zip: this zip file contains all ~680 complete assemblies.

asm_complete_mtbc.csv: the list of details for all the assemblies. Includes the country. This will be useful later for identification.

Software installation

On Ubuntu for windows (WSL) you can install most software with apt. e.g. `sudo apt install bwa`
On macOS it may be installable with brew, if not we can figure another way.